# Aaron Espinoza

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## EDUCATION

**Texas State University** **Expected Dec. 2026**
*B.S. Honors Computer Science; Minors in Applied Mathematics and Chinese* *San Marcos, TX*

- GPA 3.98/4.00 overall, 4.00/4.00 College of Science & Engineering GPA. Dean's List/President's List, all semesters.
- Extracurriculars: Vice President, TXST Fencing Club (40+ members), TXST Esports Club CS2 Captain.

## EXPERIENCE

**Open Source Contributor** **Sept. 2025 – Present**
*The FreeBSD Project* *Remote*

- Landed a cleanup in `if_bridge(4)` removing dead code, found while tracing the driver during early `veb(4)` work; reviewed and landed through the project's code review process.
- Building `veb(4)`, a virtual Ethernet bridge pseudo-device for jail and bhyve networking, modeled on OpenBSD's `veb(4)`: a stripped-down alternative to `if_bridge(4)` that stays hidden from the host network stack to cut latency.
- Maintain RustKLD, the Rust kernel driver framework delivered during GSoC, including CI and contributor docs.

**Google Summer of Code Contributor** **May 2025 – Sept. 2025**
*The FreeBSD Project* *Remote*

- Delivered a modular Rust framework for FreeBSD kernel device drivers with safe wrappers over C kernel objects, removing the need for contributors to hand-roll FFI for each new driver.
- Wrote 11 automated ATF/Kyua tests covering stress, concurrency, error handling, and memory leaks for existing drivers.
- Fixed devfs registration, quiesce handling, and LLVM 19 compatibility in the existing Rust driver, and shipped Hello World and Echo reference drivers plus documentation now used by new contributors.

**Undergraduate Deep Learning Researcher** **Aug. 2024 – Present**
*Texas State University* *San Marcos, TX*

- Research GAN-based adversarial perturbation generation against facial recognition systems using hybrid U-Net generators.
- Achieved 98.7% average impersonation success on LFW and CASIA-WebFace while keeping perturbations visually imperceptible; designed both targeted and untargeted attacks to measure model resilience.

**AI Research Intern** **May 2024 – Aug. 2024**
*Guangdong Provincial People's Hospital* *Guangzhou, China*

- Placed 1st globally in the MICCAI 2024 WHS++ whole heart segmentation challenge.
- Built ensembled PyTorch CNN segmentation models for cardiac CT and MRI, reaching a 94% Dice score.
- Applied data augmentation and model calibration to hold accuracy across imaging modalities and hospital sites.

**Maintenance Technician** **Aug. 2022 – May 2024**
*tekRESCUE* *San Marcos, TX*

- Ran weekly maintenance across client server fleets; built and configured custom servers and handled on-site troubleshooting.

## PUBLICATIONS

Enhance Multi-modal and Multi-center Whole Heart Segmentation Using Data Augmentation and Model Calibration. *CARE Workshop, MICCAI 2024.* doi.org/10.1007/978-3-031-87009-5_13
[Under review] Silent but Strong: Skip-Bottleneck Composition in Adversarial Face Impersonation. *Submitted to AAAI 2027.*

## PROJECTS

**RustKLD** github.com/Acesp25/rustkld
Framework and CI harness for writing Rust FreeBSD kernel device drivers. *Rust, C, FreeBSD, ATF/Kyua.*

**Portfolio Website**
Personal site written entirely in Rust. CSS is minified at compile time and served as a single static binary. *Rust, Rocket, Maud.*

## SKILLS

**Languages:** C, Rust, Python, Java, C++, Bash, Assembly (MIPS, x86-64); Chinese (minor, conversational)
**Systems & Networking:** FreeBSD kernel and device drivers, POSIX (pthreads, sockets, IPC), pf, WireGuard, NFS, bhyve, QEMU, jails
**Machine Learning:** PyTorch, TensorFlow, CNNs, U-Nets, GANs
**Tools:** Git, ATF/Kyua, DTrace, GDB, Wireshark, Vim, VS Code, Jira